# Answers: Introduction to logic (maths)

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**Summary**

Answers to questions relating to the guide on introduction to logic (maths).

*These are the answers to Questions: Introduction to logic (maths).*

**Please attempt the questions before reading these answers!**

## Q1 Propositions

**1.1.** "$3 < 5$" is a proposition.

**1.2.** "$5 < 3$" is a proposition.

**1.3.** "$x < 5$" is not a proposition.

**1.4.** "5 is even" is a proposition.

**1.5.** "$y$ is odd" is not a proposition.

**1.6.** "$2 \cdot 3 = 6$" is a proposition.

**1.7.** "$4 \cdot 3 = 10$" is a proposition.

**1.8.** "$2 \cdot 3 < 5x$" is not a proposition.

## Q2 Connectives

**2.1.** The logical structure of "If $x < 3$, then $y$ is even" is $P \to Q$.

**2.2.** The logical structure of "$x$ is odd and positive" is $P \wedge Q$.

**2.3.** The logical structure of "$z$ is not $-1$" is $\neg P$.

**2.4.** The logical structure of "$y$ is even or odd" is $P \vee Q$.

**2.5.** The logical structure of "$z$ is positive if and only if $y$ is negative" is $P \leftrightarrow Q$.

**2.6.** The logical structure of "If $x < 0$, then $y$ and $z$ are even" is $P \to (Q \wedge R)$.

## Q3 Truth tables

**3.1.** $P \to Q$

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| $P$ | $Q$ | $P \to Q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | T |
| F | F | T |

**3.2.** $\neg P$

| $P$ | $\neg P$ |
|---|---|
| T | F |
| F | T |

**3.3.** $(P \lor Q) \land \neg Q$

| $P$ | $Q$ | $(P \lor Q) \land \neg Q$ |
|---|---|---|
| T | T | F |
| T | F | T |
| F | T | F |
| F | F | F |

**3.4.** $\neg(P \leftrightarrow Q)$

| $P$ | $Q$ | $\neg(P \leftrightarrow Q)$ |
|---|---|---|
| T | T | F |
| T | F | T |
| F | T | T |
| F | F | F |

**3.5.** $\neg(P \land Q)$

| $P$ | $Q$ | $\neg(P \wedge Q)$ |
|---|---|---|
| T | T | F |
| T | F | T |
| F | T | T |
| F | F | T |

**3.6.** $(P \to Q) \to R$

| $P$ | $Q$ | $R$ | $(P \to Q) \to R$ |
|---|---|---|---|
| T | T | T | T |
| T | F | T | T |
| T | T | F | F |
| T | F | F | T |
| F | T | T | T |
| F | F | T | T |
| F | T | F | F |
| F | F | F | F |

**3.7.** $\neg(P \vee (Q \vee R))$

| $P$ | $Q$ | $R$ | $\neg(P \vee (Q \vee R))$ |
|---|---|---|---|
| T | T | T | F |
| T | F | T | F |
| T | T | F | F |
| T | F | F | F |
| F | T | T | F |
| F | F | T | F |
| F | T | F | F |
| F | F | F | T |

**3.8.** $(P \leftrightarrow Q) \vee (\neg R)$

| $P$ | $Q$ | $R$ | $(P \leftrightarrow Q) \vee (\neg R)$ |
|---|---|---|---|
| T | T | T | T |
| T | F | T | F |
| T | T | F | T |
| T | F | F | T |
| F | T | T | F |
| F | F | T | T |
| F | T | F | T |
| F | F | F | T |

## Q4 Equivalence

**4.1.** $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are logically equivalent. This identity is known as **De Morgan's Law**.

**4.2.** $P \to Q$ and $Q \to P$ are not logically equivalent.

**4.3.** $P \to Q$ and $(\neg P) \vee Q$ are logically equivalent.

**4.4.** $P \to Q$ and $\neg(P \vee Q)$ are not logically equivalent.

## Version history and licensing

v1.0: initial version created 03/26 by Jessica Taberner as part of a University of St Andrews VIP project.